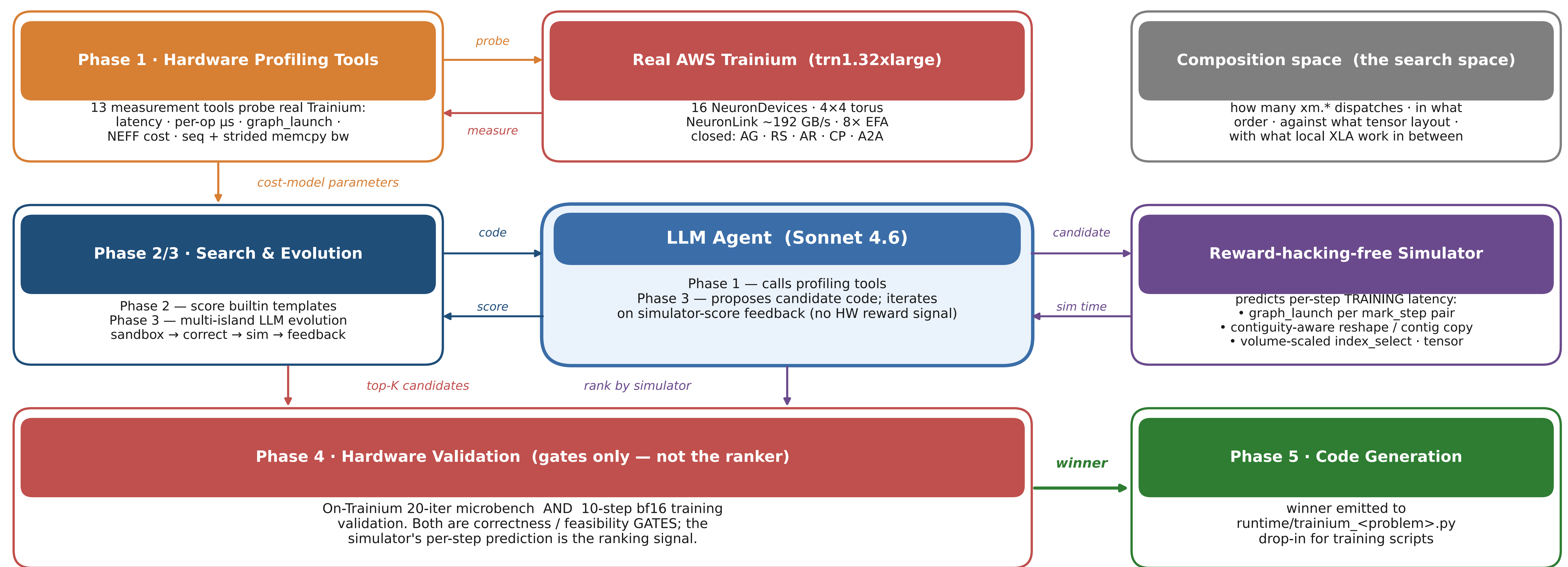


Composing Collectives Above a Black-Box Vendor Library on AWS Trainium

Profiling-driven search loop with a reward-hacking-free simulator. Hardware profiling validates correctness; the simulator predicts per-step training cost.

System overview



Why this framing

Trainium's collective library is **closed**. We can only call its primitives (xm.all_gather, xm.reduce_scatter, xm.all_reduce, xm.collective_permute, xm.all_to_all) and add local XLA ops around them. We search the **composition space** one layer up.

Prior work	Why it doesn't apply here
MSCCL / TACCL	Need open primitives + programmable schedules.
AlphaEvolve / FunSearch	Search algorithms; we compose a fixed primitive set.
AutoTVM / Triton	Search kernel impls; backend writes our kernels.

Main contributions

- Reward-hacking-free simulator**. Every cost-model term is grounded in measured device physics. Metadata-op chains, Python-built index tensors, and implicit-copy reshape patterns are priced at their actual cost — the agent cannot game the score by swapping compute ops for view ops.
- Hardware profiling only checks correctness**. The 20-iter on-Trainium microbench and 10-step bf16 training validation are gates; the simulator's per-step prediction is the ranker, because it captures NEFF cache eviction and per-mark_step overhead the isolated microbench misses.
- Contiguity-aware implicit-copy charging**. reshape/contiguous on a sub-region view is charged at strided bandwidth; on a dense permutation at sequential bandwidth. Detected from real tensor strides at trace time — never disclosed to the LLM.
- Closed-stack search loop, end-to-end validated**. All four committed runtimes match or beat the developer-written baseline at full 5000-step DeepSeek-MoE-Lite training.

Cost-model design

Per-step training latency is predicted as:

```
total_us =
  2 per_op.cost(op, copy_bytes_if_any)          -- local ops
+ 2 * graph_launch_overhead                    -- per autograd Function
+ num_dispatches * dispatch_overhead           -- per collective
+ compilation_cost(largest_collective_tensor)
  * load_events_per_run / training_steps       -- amortized NEFF reload
+ bandwidth_term(volume / link_bw)
+ cross_node_penalty
```

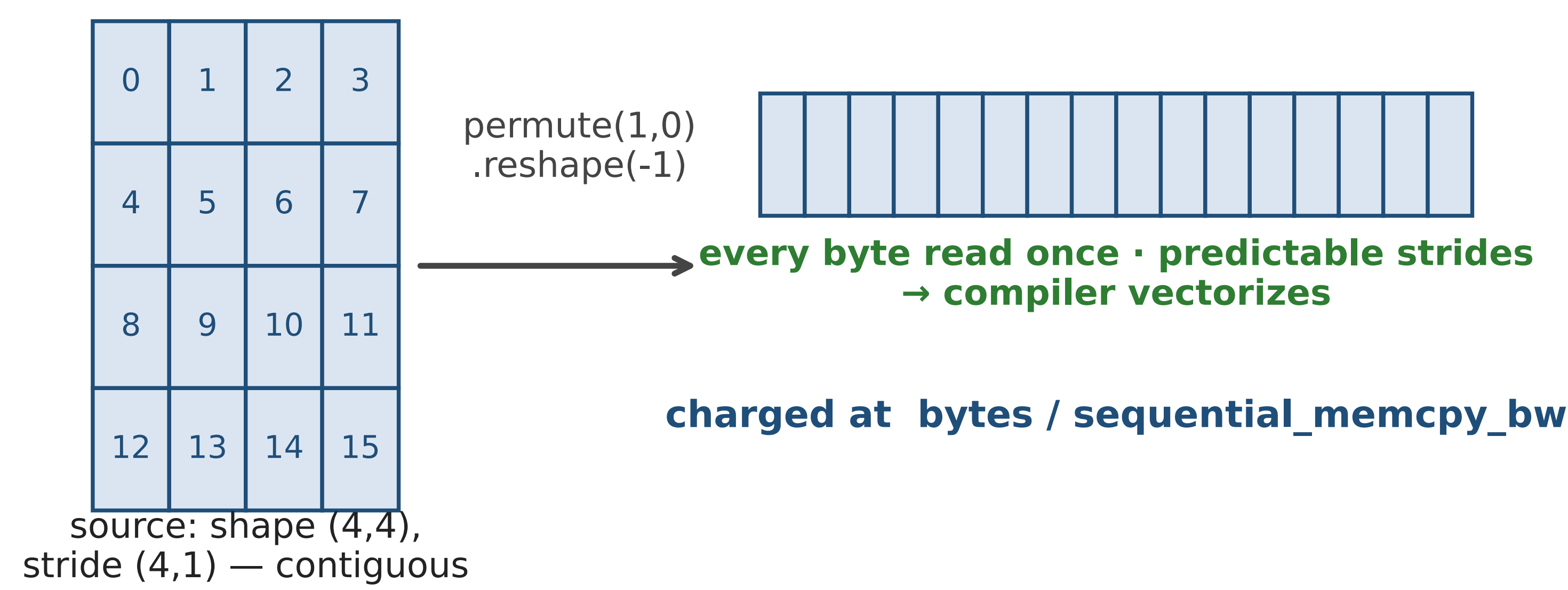
Five design choices keep the model ungameable:

- graph_launch paid **per mark_step pair**, not per collective.
- Every op has a **min_local_op_us** floor — metadata stacks can't score near zero.
- Pure view ops (view, narrow, permute, slice, ...) are floor-priced; their isolated microbench is kernel-launch overhead, not fused-graph cost.
- reshape/contiguous charge an **implicit-copy term** when strides show a non-contiguous source — two regimes, see figure.
- index_select and torch.tensor(python_list) are **volume-scaled** so a candidate can't hide an O(N) host-side build behind one flat 29 μ s cost.

Detected at trace time via PyTorch tensor strides — no leak to the agent

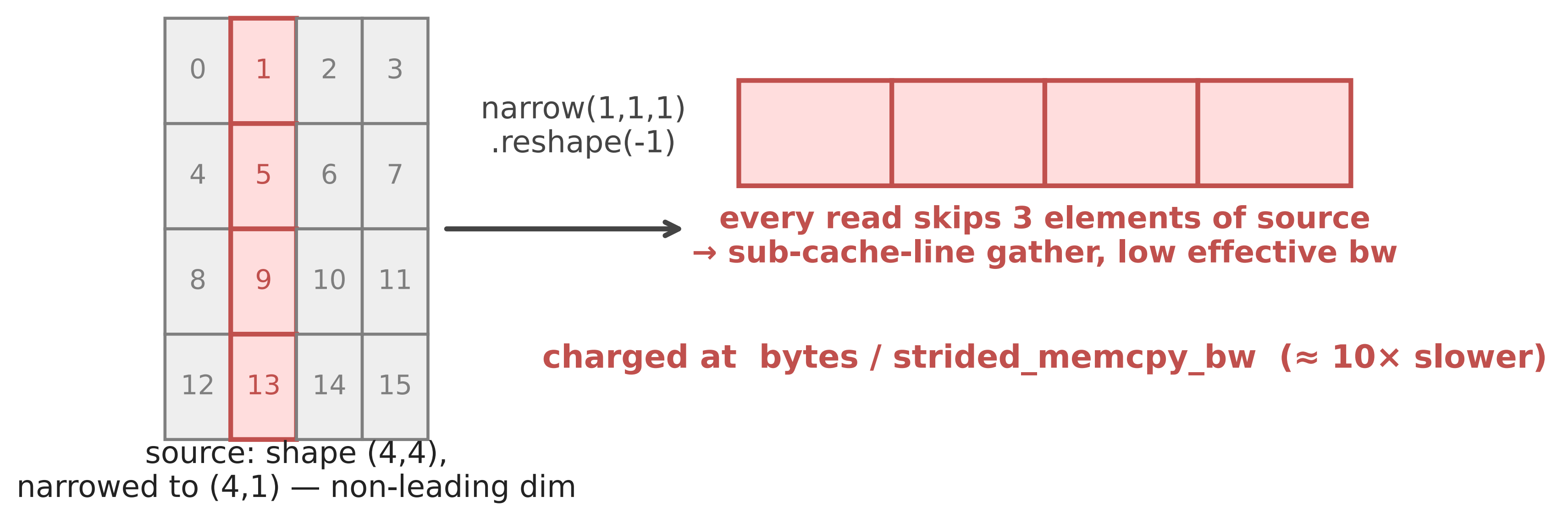
DENSE view: permute(...).reshape(-1)

source covers full storage with permuted strides



SUB-REGION view: narrow(dim=1, k, 1).reshape(-1)

source covers MORE storage than tensor needs



Dense-vs-sub-region detected from real tensor strides at trace time. Without this distinction the agent either over-rewards a narrow → reshape "trick" or mistakenly penalizes a permute → contiguous packing pattern.

